Mapping Dengue Vector Suitability In Brazil

```
knitr::opts_chunk$set(
  eval = FALSE,
  echo = TRUE,
  message = FALSE,
  warning = FALSE
)
```

Welcome

In this workshop, we will build a first species distribution model for *Aedes aegypti*, one of the major mosquito vectors associated with dengue transmission.

We will focus on Brazil because it gives us a real public health context, a national-scale map, and enough occurrence data for a meaningful teaching example.

By the end, you should be able to:

- open an RStudio project;
- download occurrence records from GBIF;
- clean and inspect coordinate data;
- load environmental predictors;
- run GLM, GBM, and Random Forest models using `biomod2`;
- build simple ensemble models;
- map predicted environmental suitability;
- explain why suitability is not the same as dengue risk.

The workshop is designed for people who are relatively new to R. Please run code slowly, one section at a time.

What Are We Modeling?

Species distribution models, often called SDMs, estimate where environmental conditions are suitable for a species.

For dengue, this can be useful because the spatial distribution of vectors such as *Aedes aegypti* helps shape where transmission could occur. But this is only one piece of the puzzle.

An SDM for *Aedes aegypti* does **not** directly model:

- human dengue cases;
- mosquito abundance;
- human movement;
- immunity;
- vector control;
- health system reporting.

It estimates environmental suitability for the vector based on the data and predictors we provide.

Open The Project

Open the file:

```
SDM_dengue_Brazil_workshop.Rproj
```

This tells RStudio that the workshop folder is the working directory. You should not need to use `setwd()`.

Run:

```
getwd()
list.files()
```

You should see folders such as `scripts`, `data`, and `outputs`.

Check Setup

Open and run:

```
scripts/01_setup_check.R
```

Or run:

```
source("scripts/01_setup_check.R")
```

If any package is missing, run:

```
source("scripts/00_install_packages.R")
```

Download Occurrence Data From GBIF

GBIF is the Global Biodiversity Information Facility. It provides public biodiversity occurrence records from museums, surveys, citizen science, and other sources.

Today, our target is:

```
target_species <- "Aedes aegypti"
target_country <- "BR"
```

Open the main workshop script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 1** and **Section 2**, then run them line by line.

The important idea is:

```
gbif_count <- rgbif::occ_count(
  scientificName = target_species,
  country = target_country,
  hasCoordinate = TRUE
)
```

Then we download the records:

```
gbif_download <- rgbif::occ_search(
  scientificName = target_species,
  country = target_country,
  hasCoordinate = TRUE,
  limit = min(gbif_count, 10000)
)

occurrences_raw <- gbif_download$data
```

If GBIF Is Slow

Use the cached backup by keeping:

```
download_from_gbif <- FALSE
```

The backup file is:

```
data/occurrences/aedes_aegypti_brazil_clean_backup.csv
```

Clean Occurrence Records

GBIF data are powerful, but they are not perfect. Records can have missing coordinates, duplicated points, or coordinates outside the study region.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 3: Clean and map occurrence records**.

The cleaning steps remove:

- missing coordinates;
- duplicate coordinates;
- coordinates outside Brazil's broad bounding box;
- zero-zero coordinates.

We also keep a workshop-sized sample so the model runs quickly.

```
max_points <- 1000
```

This is not a magic number. It is a teaching choice.

Pause And Think

Look at the occurrence map.

Ask yourself:

- Are records evenly spread across Brazil?
- Are there clusters near cities or research centers?
- Could this reflect sampling effort rather than mosquito ecology?

Load Environmental Predictors

The model needs environmental data for both presence locations and the wider study area.

For the live workshop, we use cached WorldClim bioclimatic predictors cropped to Brazil.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 4: Load environmental predictors**.

The predictors are:

- annual mean temperature;
- temperature seasonality;
- annual precipitation;
- precipitation seasonality.

These predictors are not the only possible choices. They are a simple, teachable starting point.

Run First SDMs With biomod2

Now we fit the models.

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 5: Format data and run a first biomod2 model**. Run it slowly.

The model needs:

```
species_xy <- as.matrix(occurrences[, c("decimalLongitude", "decimalLatitude")])
species_presence <- rep(1, nrow(occurrences))
```

Because we only have presence records, `biomod2` creates pseudo-absence points.

```
formatted_data <- BIOMOD_FormatingData(
  resp.name = "Aedes_aegypti",
  resp.var = species_presence,
  resp.xy = species_xy,
  expl.var = predictors,
  PA.nb.rep = 1,
  PA.nb.absences = 1000,
  PA.strategy = "random",
  filter.raster = TRUE,
  seed.val = 42
)
```

We run three example algorithms:

- GLM: Generalized Linear Model
- GBM: Generalized Boosting Model, also called boosted regression trees

- RF: Random Forest

The script also includes an options block using `biomod2::bm_ModelingOptions()`. We keep the live run on biomod2's `bigboss` defaults, but the script shows where you would change individual algorithm parameters later.

```
models_to_run <- c("GLM", "GBM", "RF")

modeling_options <- biomod2::bm_ModelingOptions(
  data.type = "binary",
  models = models_to_run,
  strategy = "bigboss",
  bm.format = formatted_data
)

model_out <- biomod2::BIOMOD_Modeling(
  bm.format = formatted_data,
  modeling.id = "first_sdm",
  models = models_to_run,
  CV.strategy = "random",
  CV.nb.rep = 1,
  CV.perc = 0.8,
  OPT.strategy = "bigboss",
  metric.eval = c("TSS", "AUCroc"),
  var.import = 1,
  nb.cpu = 1,
  seed.val = 42
)
```

What Are Pseudo-Absences?

Occurrence data tell us where the species was recorded. They usually do not tell us where the species was truly absent.

Pseudo-absences are background locations used to help the model compare known presence sites with available environmental conditions.

This is useful, but it is also a modeling choice. Different pseudo-absence strategies can change the result.

Build Ensemble Models

Ensemble models combine predictions from the individual models.

In the workshop script, we build two simple ensemble examples:

- `EMmean`: average prediction across selected models
- `EMwmean`: weighted average, where better-performing models get more weight

The key function is:

```
ensemble_out <- biomod2::BIOMOD_EnsembleModeling(
  bm.mod = model_out,
  models.chosen = "all",
  em.by = "all",
  em.algo = c("EMmean", "EMwmean"),
  metric.select = "all",
  metric.eval = c("TSS", "AUCroc"),
  var.import = 1
)
```

Project Suitability Across Brazil

After fitting the individual and ensemble models, we predict suitability across the whole study area:

```
projection_out <- biomod2::BIOMOD_Projection(
  bm.mod = model_out,
  proj.name = "Brazil_current",
  new.env = predictors,
  models.chosen = "all"
)

ensemble_projection_out <- biomod2::BIOMOD_EnsembleForecasting(
  bm.em = ensemble_out,
  bm.proj = projection_out,
  models.chosen = "all"
)
```

The outputs are saved as:

```
outputs/maps/aedes_aegypti_brazil_suitability.tif
outputs/maps/aedes_aegypti_brazil_ensemble_suitability.tif
```

Map And Interpret

Stay in the same script:

```
scripts/workshop_aedes_aegypti_brazil_sdm.R
```

Find **Section 7: Project and map suitability across Brazil** and **Section 8: Interpret the model output**.

This creates:

```
outputs/maps/aedes_aegypti_brazil_suitability.png
outputs/maps/aedes_aegypti_brazil_ensemble_suitability.png
```

Interpretation Questions

Discuss:

- Where does the model predict high suitability?
- Does that match what you know about *Aedes aegypti*?
- Where might the map be influenced by sampling bias?
- What does the model miss?
- What would you need before using this in a public health decision?

Adapting The Workflow

To model another species, change:

```
target_species <- "Aedes albopictus"
target_country <- "BR"
```

Or choose another country code:

```
target_country <- "ZA"
```

You would also need environmental predictors that cover the new study area.

What To Remember

An SDM is a useful way to connect occurrence data and environmental predictors.

For infectious disease work, SDMs are most powerful when combined with:

- vector biology;
- surveillance data;
- human population data;
- uncertainty checks;
- local epidemiological knowledge.

The map is a starting point for questions, not the final answer.